INCIDENT RESPONSE PLAYBOOK

Disclaimer and Scope of Use

This Incident Response Playbook outlines proposed procedures designed for the GMDC scenario. The screenshots included serve as documentation of the lab setup and the Wazuh dashboard at the time of capture. They should not be interpreted as evidence of an actual cyberattack or confirmation that a complete incident-response exercise was conducted.

Purpose, Scope and Objectives

This playbook provides a repeatable process for responding to suspected or confirmed cybersecurity incidents affecting GMDC information systems. It is designed to support confidentiality, integrity, availability, reliable incident communication, and documented decision-making.

Scope

GMDC-style user endpoints, servers, identity and access systems, network services, web applications, data repositories, security monitoring, and cloud-hosted monitoring infrastructure. Any action affecting operational technology or safety-related services must be coordinated with the responsible operational owner.

Objectives

- Detect, validate, classify, and record suspected incidents.
- Contain threats proportionately while protecting essential services and evidence.
- Remove the cause of compromise and restore systems safely.
- Communicate decisions to the correct technical and management contacts.
- Use lessons learned to improve controls, monitoring, training, and risk treatment.

Incident Response Lifecycle

Stage	Key Action	Expected record
Prepare	Maintain asset inventory, contact list, backups, approved access, logging, and response procedures.	Readiness checklist; roles and contacts
Detect and identify	Review Wazuh alerts and system/network logs; verify timestamps, affected host, account, and activity.	Alert reference; evidence notes
Classify and escalate	Assign Low, Medium, High, or Critical severity based on credible evidence and potential impact.	Severity and rationale; notification log
Contain	Limit attacker access, isolate an affected endpoint, disable compromised accounts, or block traffic where appropriate.	Action, time, approver, and impact
Eradicate	Remove malicious files/persistence, patch exploited weaknesses, rotate exposed credentials, and validate configuration.	Changes made; validation results
Recover	Restore from trusted backups or rebuild, reconnect cautiously, and monitor for recurrence.	Recovery checks and owner approval
Review and improve	Document root cause, impact, response quality, and corrective actions.	Incident report; lessons learned

Decision rule: if an event is not confirmed as an incident, document the checks performed and the reason for that decision.

Keep monitoring if uncertainty remains, and reopen the case if new evidence appears.

Incident Types and Initial Indicators

Incident type	Possible indicators	Initial considerations
Malware / ransomware	Endpoint detection, unusual processes, suspicious files, unexpected encryption or service disruption.	Isolate affected endpoint if safe; preserve alert and endpoint details; check for spread.
Unauthorized access	Successful login at an unusual time/location, privilege changes, unknown account, repeated authentication failures followed by success.	Validate the account and login; disable/restrict access if compromise is credible.
Data breach	Unexpected data transfer, unusual database queries, public exposure, unauthorized downloads.	Determine data type and scope; preserve logs; notify the data owner and follow applicable reporting procedures.
Compromised server / backdoor	Unexpected listening service, suspicious process, altered system files, unusual privileged activity.	Treat as high risk; coordinate containment and evidence collection with the server owner.
Network intrusion / scanning	Unexpected connections, repeated probes, unusual outbound traffic or security-device alerts.	Identify source, target, scope and whether access succeeded; block only with appropriate approval.
Misconfiguration / exposed service	Directory listing, outdated software, excessive permissions, insecure remote service.	Record as a vulnerability unless evidence indicates active exploitation or unauthorized activity.

Incident Severity and Escalation

Level	Guideline	Initial response
Critical	Mission-critical service compromised/disrupted; active destructive malware; confirmed sensitive-data exfiltration; privileged takeover with broad impact.	Immediate escalation to Incident Lead and senior management; activate response team; urgent coordinated containment.
High	Credible successful unauthorized access or exploitation of an important server, account, or data repository; serious service impact.	Prompt escalation to Incident Lead and relevant asset/data owner; assign responders and preserve evidence.
Medium	Suspicious event requiring investigation, limited unauthorized access, or contained malware with no current evidence of major impact.	Investigate during the current working period; record findings and raise severity if evidence worsens.
Low	Isolated low-impact event with no evidence of successful compromise or material disruption.	Log, validate, correct if needed, and monitor for recurrence.

These are proposed internal categories and response priorities. The incident lead may increase or decrease severity as evidence and impact become clearer. A vulnerability scan finding alone is not automatically a security incident.

Roles and Responsibilities

Role	Responsibilities
Incident Lead / Security Owner	Owns incident coordination, classification, escalation, decisions, incident record, and closure approval.
SOC Analyst / First Responder	Reviews monitoring alerts, validates evidence, records timelines, and recommends severity and next steps.
System Administrator	Investigates affected hosts, isolates or rebuilds systems when approved, patches weaknesses, and verifies recovery.
Network Administrator	Reviews network evidence and applies approved network containment or filtering.
Asset / Data Owner	Explains business importance and data sensitivity; approves operationally significant changes and recovery.
Management	Provides authority and resources for major incidents and receives status updates for High/Critical cases.
Authorized Users	Report suspicious activity promptly and do not delete evidence or investigate beyond their authorization.

Response Procedure

1. Detect and report

Capture the alert or report, timestamp, source, affected host/account, and initial observations. Create an incident ID.

2. Validate and identify

Check whether the alert is plausible. Correlate relevant logs, endpoint activity, authentication records, and network evidence.

Record what is known and what remains uncertain.

3. Classify and escalate

Assign severity using the classification table. Notify the Incident Lead and the asset/data owner according to severity. Record the rationale and notifications.

4. Contain

Choose proportionate actions such as isolating an endpoint, disabling a compromised account, restricting a service, or blocking a source. Record who approved the action and its operational impact.

5. Eradicate

Remove confirmed malicious components, close the exploited path, patch or reconfigure affected systems, rotate exposed credentials, and check for persistence or related compromise.

6. Recover

Restore from trusted backups or rebuild where necessary. Verify service function, security controls, account integrity, and monitoring before returning the asset to normal use.

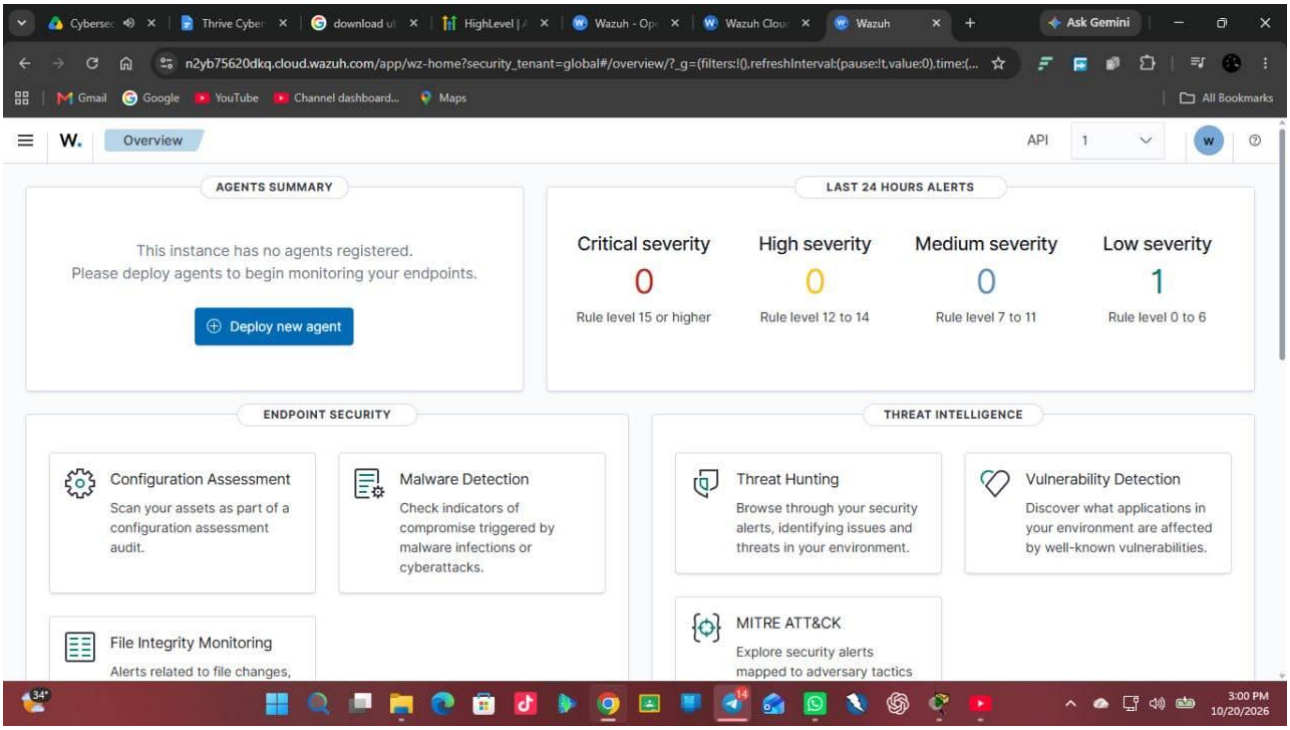
7. Review and close

Document timeline, scope, root cause where known, impact, actions, evidence references, recovery checks, lessons learned, and follow-up owners. Close only after approval.

Wazuh SOC Lab Setup and Implementation Evidence

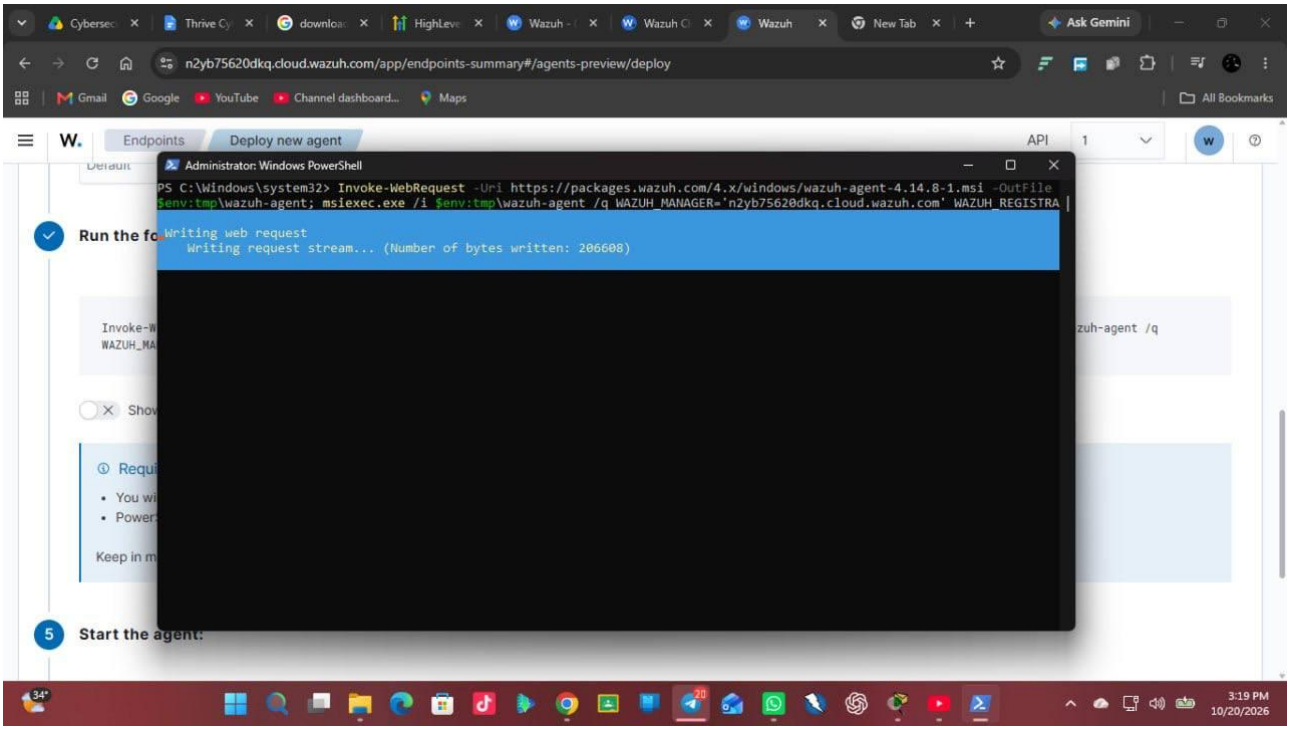
The evidence supplied for this assignment shows a cloud-hosted Wazuh dashboard and Windows endpoint-agent deployment activity.

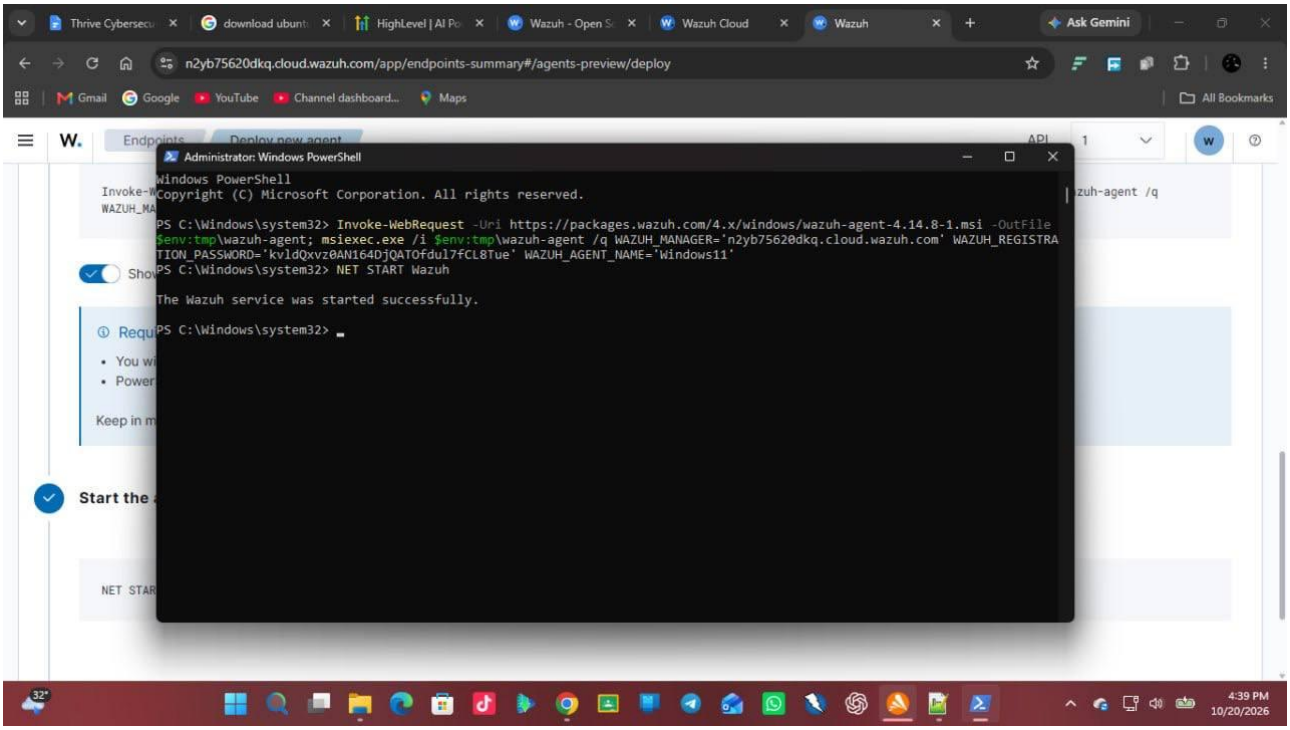
Initial dashboard state



Initial Wazuh overview dashboard before an agent was registered.

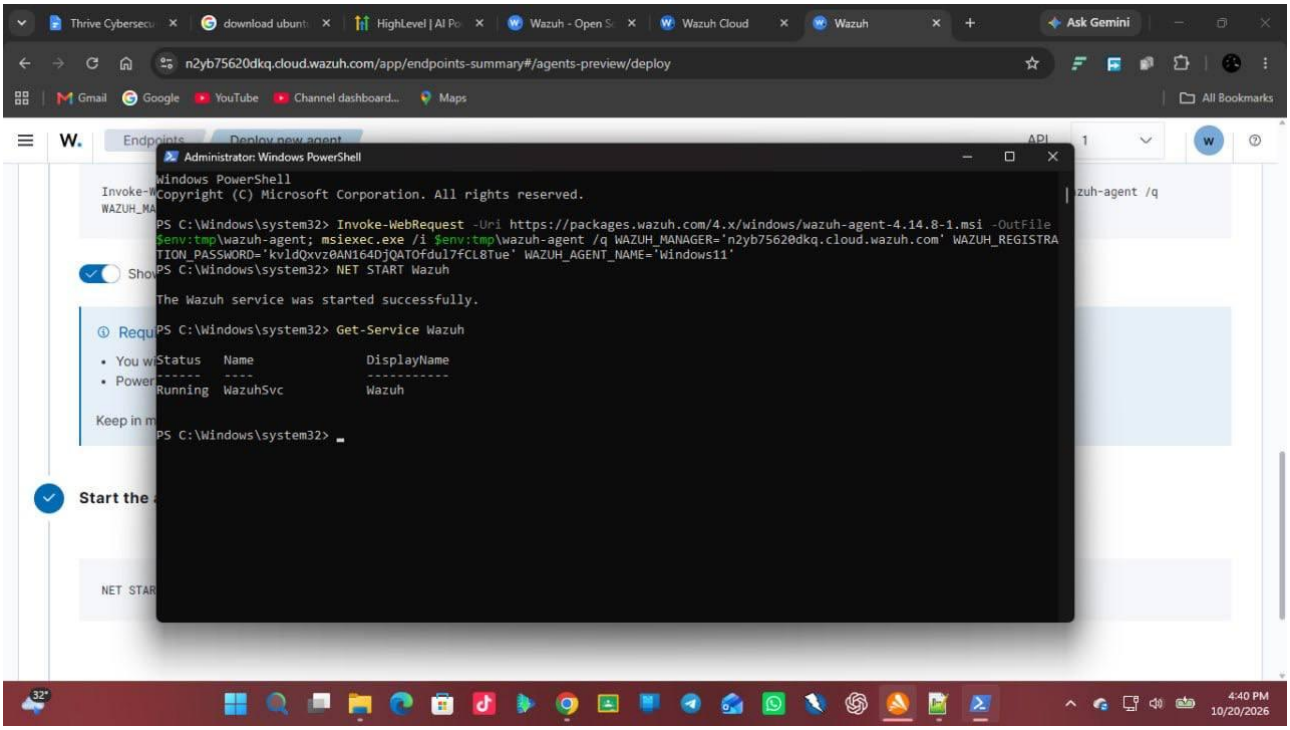
Agent deployment command





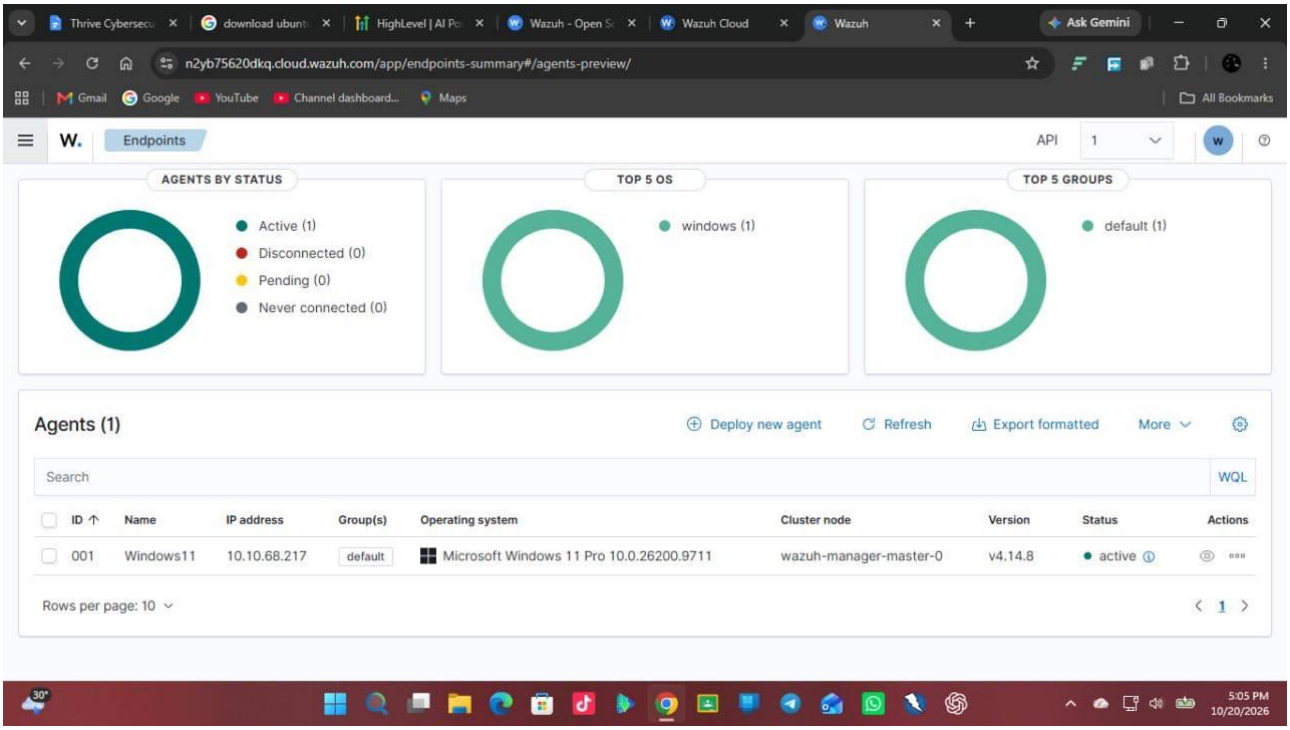
PowerShell command used during Windows endpoint-agent deployment.

Wazuh service start/status



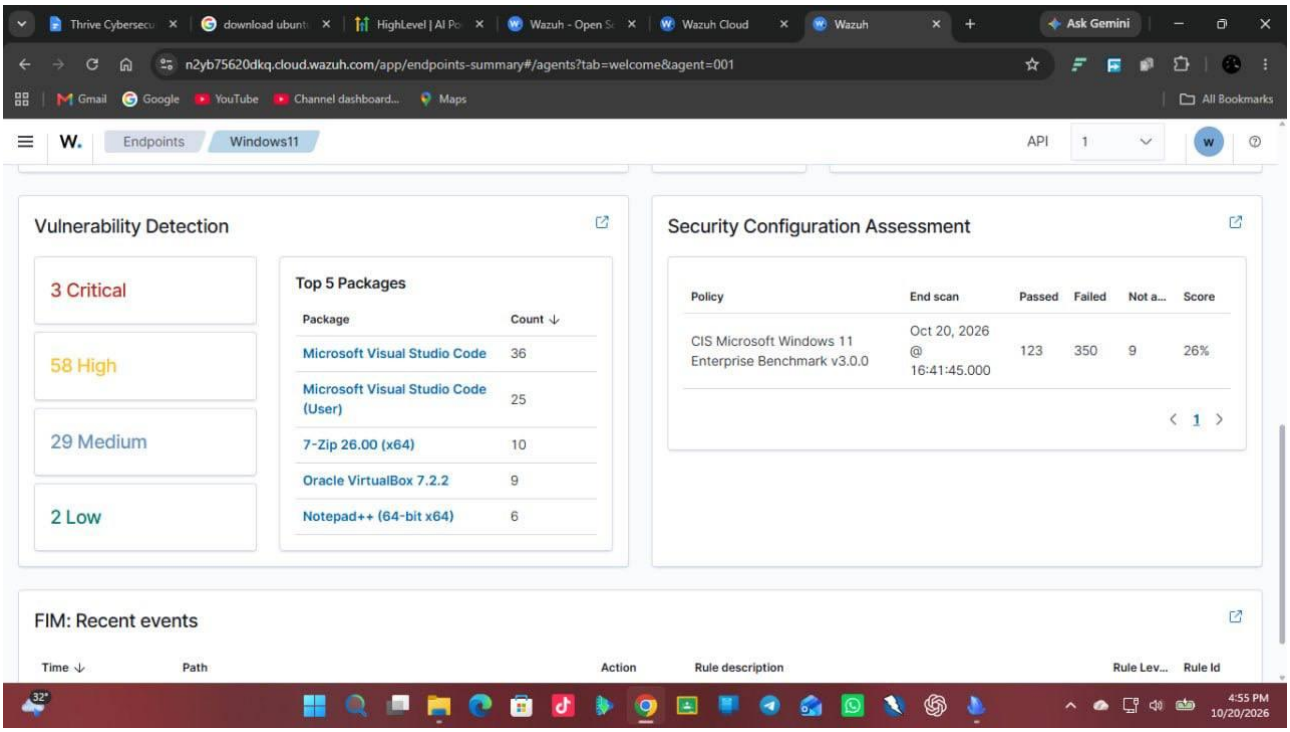
PowerShell output showing the Wazuh agent service start/status check.

Agent shown as active



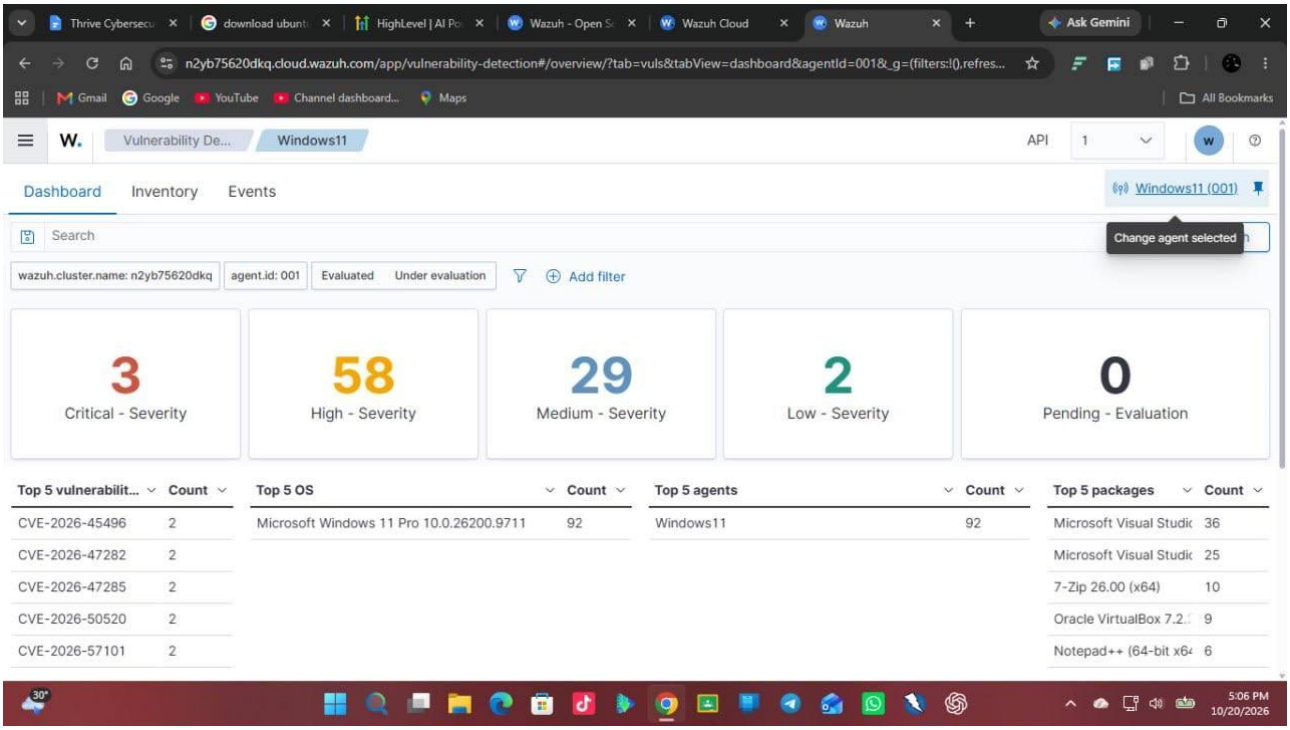
Wazuh Agents view showing the Windows11 endpoint with Active status at the time of capture.

Endpoint vulnerability and configuration overview



Wazuh endpoint view showing vulnerability-detection counts and a security configuration assessment summary.

Vulnerability detection summary



Wazuh vulnerability-detection dashboard showing severity totals for the selected Windows endpoint.

How to Use Wazuh During an Incident

Alert review

Open the relevant Wazuh alert or module, note its timestamp, rule/severity, affected agent, and description. Do not treat every alert as a confirmed compromise.

Endpoint context

Confirm that the alert relates to the correct endpoint and compare it with the asset owner, operating system, recent changes, and other relevant events.

Evidence collection

Export or record relevant alert details where permitted. Keep original timestamps and record who collected the evidence and when.

Triage and classification

Use the incident classification table to determine Low/Medium/High/Critical. Escalate if evidence suggests privileged compromise, sensitive-data exposure, or disruption to critical operations.

Response and verification

Coordinate containment, eradication, and recovery with the appropriate owner. After recovery, review monitoring for recurrence and record the result.

Communication and Escalation

Severity	Notify	Communication expectations
Critical	Incident Lead, senior management, affected system/data owner, relevant technical responders.	Immediate escalation; provide concise updates at agreed intervals and when impact or containment status changes.
High	Incident Lead and relevant system/data owner; management according to internal rules.	Prompt escalation; update at major response milestones.
Medium	Security/IT response team and affected asset owner.	Record the case and communicate findings/actions during investigation.
Low	Security/IT queue or designated owner.	Record, correct if needed, and close with rationale.

Communication should be factual and need-to-know. Avoid sending passwords, tokens, private keys, or unnecessary personal data in incident updates. Use approved channels and follow any applicable privacy, legal, or regulatory process.

Evidence Handling and Incident Record

- Preserve original logs and files where possible; work on copies for analysis when appropriate.
- Record evidence source, collection time, collector, host/account, and any transfer or storage location.
- Restrict access to evidence and protect it from modification or unauthorized disclosure.
- Record actions and decisions in chronological order, including approvals and service impact.

Do not attempt unauthorized testing or disruptive containment against production or operational technology systems.

Incident field	**What to record**
Incident ID / timestamps	Unique identifier, detection time, report time, and timezone.
Affected asset / owner	Hostname or IP, account, service, data type, and responsible owner.
Evidence / alert details	Wazuh rule/alert reference, log source, relevant timestamps, and evidence location.
Severity / rationale	Assigned level and factual basis; note uncertainties.
Actions / approvals	Containment, eradication, recovery steps, who approved them, and when.
Outcome / follow-up	Impact, recovery validation, lessons learned, action owner, and closure approval.

Recovery, Closure and Continuous Improvement

- Confirm the system is stable, required services function, accounts and permissions are correct, and relevant security monitoring is active.
- Check for recurrence of the same alert, suspicious account activity, unexpected processes, or unusual network connections.
- Document known root cause and scope; if these are unknown, state that clearly rather than guessing.
- Assign corrective actions such as patching, configuration hardening, credential rotation, improved logging, or user training.
- Review the playbook after a significant incident, a material system change, or a scheduled governance review.

Conclusion

This Incident Response Playbook provides a structured approach for identifying, classifying, containing, eradicating, and recovering from cybersecurity incidents within the Ghana Maritime Defense Command (GMDC) scenario. It establishes clear responsibilities, escalation procedures, communication guidelines, and recovery processes to support an effective and coordinated response to potential security threats.

The integration of Wazuh into the proposed security monitoring process demonstrates how a Security Information and Event Management (SIEM) platform can support endpoint monitoring, security event analysis, and incident investigation. The accompanying screenshots document aspects of the Wazuh lab setup and dashboard status; they are evidence of the setup process and should not be interpreted as proof of a confirmed cyberattack.

Ultimately, the effectiveness of this playbook depends on regular reviews, appropriate staff training, timely incident reporting, and continuous improvement of security controls. By following these procedures, GMDC can strengthen its incident response readiness and improve its ability to protect information assets and support the continuity of essential operations.